Supply Chain Black Box: Hidden Supplier Dependency Discovery, Calibrated Cascading-Risk Prediction and Optimised Recovery for an FMCG Manufacturer

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***Abstract*—A chocolate and malt-beverage manufacturer sees its Tier-1 suppliers but not the Tier-2/3 suppliers they share, so one hidden trader can starve several "independent" cocoa processors at once. We build and evaluate an end-to-end decision system that (i) predicts 30-day supplier disruption with calibrated probabilities using point-in-time features, (ii) ranks candidate hidden relationships by link prediction, (iii) propagates disruptions through a daily material-flow simulator, and (iv) converts predictions into cost-minimising actions and a mixed-integer recovery plan. Supplier-level experiments use a documented synthetic FMCG ecosystem (500 suppliers, 1,500 relationships, of which 69 % are hidden); two real public datasets test transferable components (DataCo, 172,765 shipments; SupplyGraph, a real FMCG graph). Platt scaling halves XGBoost's expected calibration error (0.020→0.010); the observed network understates Tier-2 and Tier-3 exposure by 2.1× and 10.8×; GNNs did *not* outperform feature-based models on either synthetic or real data. In a predefined nine-scenario evaluation (500 common-random-number replications), optimisation-based recovery reduces the simulated impact of a severe cocoa-trader outage from INR 4.72 to 1.16 crore (75 % lower) and is also cheapest under a moderate shock, but over-hedges when the shock is mild.**

***Index Terms*—supply chain risk, link prediction, graph neural networks, probability calibration, cascading disruption, recovery optimisation, FMCG**

I. Introduction

A. Business Problem

Disruptions are costly: firms announcing supply chain disruptions experienced significantly lower long-run stock returns and higher equity risk [1]. In FMCG the exposure is structural: a malt-beverage jar needs malt extract, milk powder, cocoa, sugar, vitamin premix and PET jars, and each Tier-1 input depends on Tier-2 commodity traders that procurement rarely maps. Disruptions therefore "ripple" through invisible links [2]. The business decision we support is concrete: *for each supplier, each month, which mitigation action should procurement take, and if a shock occurs, which alternative suppliers should receive which quantities by which transport mode?*

B. Scope, Objectives and Contributions

Objectives: discover candidate hidden dependencies; predict calibrated disruption probabilities; quantify cascading inventory, production and financial exposure; and optimise recovery. Contributions: (1) an integrated, executable pipeline in which every recommendation is traceable to a model output, an exposure calculation and a solver result; (2) a leakage-audited evaluation (prediction timestamp, embargoed chronological splits, calibration fitted only on a calibration split, test edges removed from message passing); (3) a quantification of the "black-box gap" between observed and true network exposure; (4) a predefined nine-scenario policy comparison with explicit economics; (5) a tested React/Flask/MySQL application. We do not claim novelty for GNNs, calibration or MILP individually.

II. Related Work

Risk prediction. AI for supply chain risk management is dominated by risk identification and assessment, with comparatively few studies closing the loop to response decisions [3]. Gradient boosting [4] is a strong tabular baseline. Because actions are triggered by probabilities, calibration matters: Platt scaling [5], isotonic regression [6] and their comparison across learners [7]. **Graph-based intelligence.** Brintrup *et al.* [8] framed hidden-link discovery as link prediction on a real automotive network; Kosasih and Brintrup [9] used GNNs for the same task. Classical heuristics [10], node2vec [11] and message-passing models GCN [12], GraphSAGE [13] and GAT [14] are our comparators; SupplyGraph [15] provides a real FMCG benchmark. **Cascades and recovery.** OR/MS research models disruptions with inventory, sourcing and contracting decisions [16], and the ripple-effect literature studies propagation [2]. **Gap.** Prior work treats discovery, prediction and recovery separately and rarely reports calibrated, leakage-controlled, policy-level business outcomes; we evaluate their combination with explicit economics.

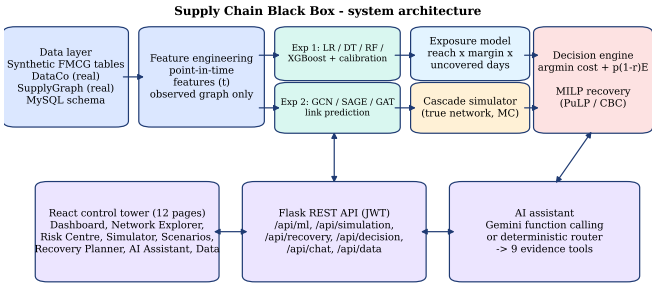


Fig. 1. System architecture: data layer, point-in-time features, prediction and link models, exposure model, cascade simulator, decision engine with MILP, Flask API, React control tower and tool-calling assistant.

III. Methodology

A. Data and Provenance

Synthetic FMCG ecosystem (fictional "Cocoa & Malt Foods", seed 42): 120/200/180 Tier-1/2/3 suppliers, 50 materials over three BOM levels, 100 products, two plants, 1,048 disruptions (782 qualifying) over 2024–2025, 4,929 POs, 9,579 shipments, 5,824 weekly inventory records. A buyer needing material m picks $k \in \{2..6\}$ sources with probability $\propto$ size$^{1.6} e^{-d/2500\,\mathrm{km}}$; Tier-2 and Tier-3 edges are observed with probability 0.45 and 0.15, so 461 of 1,500 relationships are visible. Daily disruption hazard combines latent financial risk, utilisation, regional geo-risk, monsoon/cyclone seasonality, an unobserved frailty and contagion from disrupted upstream suppliers over the *true* network. Every row carries a synthetic label; hidden edges are used only as simulator ground truth and for evaluation. **DataCo** [17] (CC BY 4.0; GitHub mirror, SHA-256 recorded): 180,519 order lines, 7,754 cancelled removed. **SupplyGraph** [15]: 40 products (one duplicated node removed), 684 product-pair edges, 221 days. Real and synthetic identities are never merged. All tables pass 196 validation checks (keys, FKs, dates, ranges, logical rules).

B. Disruption Prediction and Leakage Control

The prediction timestamp t is 00:00 on the first of each month. For supplier i,

$$y_i(t) = \mathbb{1}\big[\exists e:\ s_e \in (t, t+30\mathrm{d}],\ \ell_e \geq 0.25,\ d_e \geq 3\big], \quad (1)$$

with start s_e, capacity loss ℓ_e and duration d_e. The 25 features $x_i(t)$ use only disruptions with reported date $< t$, deliveries completed before t, the monthly risk forecast for t and *observed* edges (e.g., upstream disruptions, single sourcing, delay and fill rate over 90 days). Splits: train 2024-04..2025-02 (5,500 rows), one-month embargo, calibration 2025-04..06 (1,500), test 2025-07..12 (3,000). Logistic regression, a depth-5 tree, random forest and XGBoost are fitted on train only.

C. Probability Calibration

Platt scaling fits $q = \sigma(a\,\mathrm{logit}(\hat{p}) + b)$ and isotonic regression a monotone map, both on the calibration split; the method is chosen by 5-fold CV Brier score within that split. We report Brier $= \frac{1}{N}\sum(q_i - y_i)^2$ [7] and ECE $= \sum_b \frac{n_b}{N}\,|\bar{y}_b - \bar{q}_b|$ with 10 equal-width bins.

D. Graph Learning

$G = (V, E)$ has 501 nodes (suppliers and manufacturer) with 131 observable attributes (tier, output and input materials, region, size, reliability, financial risk, utilisation). GraphSAGE updates $h_v^{(l+1)} = \mathrm{ReLU}(W_1 h_v^{(l)} + W_2\,\mathrm{mean}_{u\in N(v)} h_u^{(l)})$; a pair decoder scores $\sigma(\mathrm{MLP}[z_u \odot z_v, |z_u - z_v|])$. **A1**: observed edges split 70/10/20, test edges removed from the message-passing graph, negatives random or *material-compatible*. **A2**: train on all observed edges and rank all 8,863 compatible unobserved pairs against 1,039 hidden edges. The model used downstream is selected by validation AP only. **Exp. 2B** feeds Exp. 1 features through GCN/GraphSAGE/GAT on observed, observed+top-500 predicted, and oracle graphs, and on SupplyGraph predicts a 7-day service shortfall (deliveries $< 80\,\%$ of orders) with a 7-day-embargoed 60/20/20 date split.

E. Exposure, Cascade Simulation and Decision Rule

Dependency concentration of material m is $\mathrm{HHI}_m = \sum_s w_{sm}^2$. Supplier i's reach r_{im} multiplies dependency shares along paths to the manufacturer (divided by the buyer's number of inputs). Exposure is

$$E_i = \sum_m r_{im}\,\mathrm{CM}_m\,\mathbb{E}\big[S\max(0, D - c_m - 10(\tau_i - 1))\big], \quad (2)$$

with daily contribution margin CM_m of products using m, cover c_m (days), tier τ_i and empirical severity/duration (S, D). The simulator propagates availability daily over the true network:

$$a_b(t) = \min\big(1 - \ell_b(t),\ \min_i \min(1,\ A_{bi}(t))\big), \quad (3)$$

where $A_{bi}(t) = I_{bi}(t) + \sum_u w_{ub}\,a_u(t - L_{ub})$ is the available input i in days of need, 10-day input buffers I_{bi}, lead times L_{ub} and a BOM-bottleneck production rule at the plants; unmet demand is $D_p(1 - \min_{m\in\mathrm{BOM}(p)} \mathrm{fill}_m)$. With calibrated q_i, the executable action is

$$a^* = \arg\min_a\ C_a + q_i(1 - \rho_a)E_i, \quad (4)$$

over monitoring, enhanced monitoring, contingency review and escalation (C_a=0/5k/20k/60k INR, mitigation ρ_a=0/0.15/0.45/0.70), which is Bayes-optimal under calibration and yields cost-derived thresholds $q^* = \Delta C/(E_i \Delta\rho)$.

F. Recovery Optimisation and Nine-Scenario Design

For deficits δ_m, the MILP (PuLP/CBC) chooses quantities x_{jk} from candidate j (approved or substitute supplier, compatibility κ_j) by mode k (standard/expedited), onboarding $z_j \in \{0, 1\}$ and shortage s_m:

$$\min\ \textstyle\sum_{jk} p_j(1 + f_{jk} + hW/2)x_{jk} + \sum_j F z_j + \sum_m \pi_m s_m \quad (5)$$

$$\text{s.t. } \textstyle\sum_{jk} \rho_j(1 - q_j)\kappa_j\alpha_j x_{jk} + s_m \geq \delta_m,$$
$$x_{jk} \leq u_j(W - L_{jk})^+,\ \ \textstyle\sum_k x_{jk} \leq 0.6\,\delta_m,$$
$$x_{jk} = 0 \text{ if } q_j > 0.35,\ \ \textstyle\sum_k x_{jk} \leq M z_j,$$

TABLE I

EXP. 1: 30-DAY SUPPLIER DISRUPTION PREDICTION ON THE UNTOUCHED TEST SPLIT (3,000 SUPPLIER-MONTHS, PREVALENCE 7.4%). CALIBRATORS FITTED ON THE CALIBRATION SPLIT ONLY; SHADED = DEPLOYED.

Model	Calib.	ROC-AUC	PR-AUC	Brier	ECE_{10}
Logistic reg.	–	0.666	0.150	0.0669	0.0066
Logistic reg.	Platt	0.666	0.150	0.0670	0.0059
Logistic reg.	Iso.	0.659	0.126	0.0687	0.0070
Decision Tree	–	0.571	0.098	0.0704	0.0366
Decision Tree	Platt	0.571	0.098	0.0689	0.0111
Decision Tree	Iso.	0.569	0.094	0.0689	0.0124
Random Forest	–	0.657	0.140	0.0672	0.0046
Random Forest	Platt	0.657	0.140	0.0675	0.0081
Random Forest	Iso.	0.655	0.124	0.0673	0.0063
XGBoost	–	0.626	0.122	0.0689	0.0197
XGBoost	Platt	0.626	0.122	0.0680	0.0104
XGBoost	Iso.	0.620	0.113	0.0681	0.0109

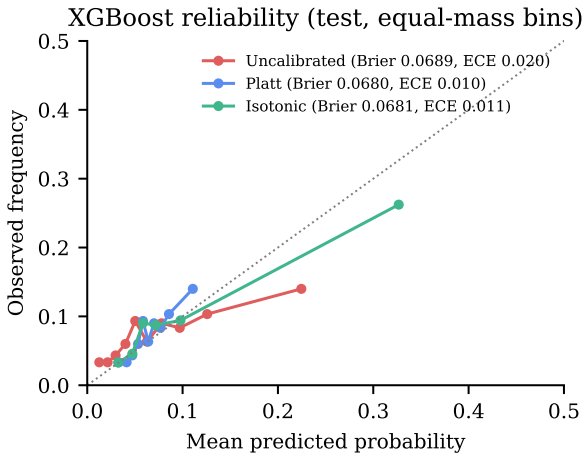


Fig. 2. XGBoost reliability on the test split before and after calibration (calibrators fitted on the calibration split).

TABLE II

EXP. 2A LINK PREDICTION (SYNTHETIC, 5 SEEDS FOR LEARNED MODELS). A1: HELD-OUT OBSERVED EDGES VS. MATERIAL-COMPATIBLE NEGATIVES. A2: RANKING ALL 8,863 COMPATIBLE UNOBSERVED PAIRS AGAINST 1,039 GROUND-TRUTH HIDDEN EDGES (BASE RATE 0.117).

	A1 (held-out)		A2 (hidden-edge discovery)		
Method	ROC-AUC	AP	AP	P@100	R@500
Common neigh.	0.500	0.500	0.117	0.06	0.042
Adamic-Adar	0.500	0.500	0.117	0.06	0.042
Pref. attach.	0.733	0.763	0.166	0.34	0.120
Node2Vec	0.638	0.609	0.146	0.24	0.084
MLP	0.826±0.017	0.849±0.012	0.227	0.32	0.141
GCN	0.807±0.020	0.837±0.010	0.176	0.25	0.109
GraphSAGE	0.737±0.009	0.784±0.012	0.168	0.29	0.102
GAT	0.796±0.005	0.811±0.018	0.177	0.26	0.103
GraphSAGE+skip	0.792±0.014	0.827±0.006	0.166	0.29	0.102

with spare capacity u_j, deadline W, freight f_{jk}, holding rate h, shortage penalty π_m and availability α_j derated by the candidate's estimated dependency on the disrupted suppliers. The shock disrupts the five largest Tier-2 cocoa-bean traders (19 observed and 25 hidden outgoing links) by 25/50/100 % for a nominal 60 days. Policies: A none; B rule-based (observed graph, same-material alternates by reliability, standard freight, reactive trigger at 10 days of cover); C optimisation (observed+predicted graph, MILP at detection plus the same reactive trigger). All policies share network, demand, inventory and economics; replication r uses seed $s_0 + r$ for every policy (common random numbers) over 180 days. Illustrative economics (INR): premium 6 % (10 % substitute), onboarding 1.5 lakh, freight 2.5/4 % (expedited 3×), holding 24 %/yr, shortage penalty 15 % of price, interruption 75k per plant-day, recovery management 25k/day. Impact is the sum of lost contribution margin, penalties, procurement premium, transport, holding, onboarding, interruption and management costs.

IV. RESULTS

A. Disruption Prediction and Calibration

Table I: discrimination is modest, as expected for rare, partly contagion-driven events (prevalence 7.4 %); logistic regression ranks best (ROC-AUC 0.666, PR-AUC 0.150) and the deployed XGBoost reaches 0.626/0.122. Calibration matters for decisions: Platt reduces ECE from 0.020 to 0.010 for XGBoost and from 0.037 to 0.011 for the tree (Fig. 2). SHAP attributes most signal to utilisation, the weather forecast and financial risk. On the test months the cost-based policy issued 5 contingency alerts (0 true positives); realised cost was 118.3 lakh versus 116.9 (monitor only), 120.4 (uncalibrated) and 664.3 (review all), with a perfect-foresight bound of 82.1. Because 3–4 weeks of inventory absorb most single-supplier events, per-supplier exposures rarely justify action—an honest negative result that motivates network-level response.

B. Graph Learning and the Black-Box Gap

With only observed links, mean exposure of Tier-2 and Tier-3 suppliers is understated 2.1× and 10.8× relative to the true network. For discovery (Table II), common neighbours and Adamic–Adar are uninformative on the layered network (no shared neighbours); the validation-selected MLP pair model attains AP 0.227 against a base rate of 0.117 (P@100 = 0.32), whereas GraphSAGE reaches 0.168. Table III shows that GNNs did not improve risk classification over tabular models on the synthetic network—even with the oracle graph—or on real SupplyGraph, where logistic regression is best. We therefore treat candidate links as verification leads, not as confirmed relationships.

C. Real-Data Shipment Delay (DataCo)

On 34,554 chronologically held-out order lines, XGBoost reaches ROC-AUC 0.780 (PR-AUC 0.858) versus 0.744 for a shipping-mode baseline (Table IV); delay-days regression adds nothing over the mode mean (R^2 0.27 vs. 0.28).

D. Nine-Scenario Business Evaluation

Table V and Fig. 3 report all thirteen required outcomes. Mild shocks are absorbed by buffers for every policy. Under the moderate shock, impacts are 61.5 (A), 50.5 (B) and 44.8 lakh (C). Under the severe shock, no intervention leaves 734k

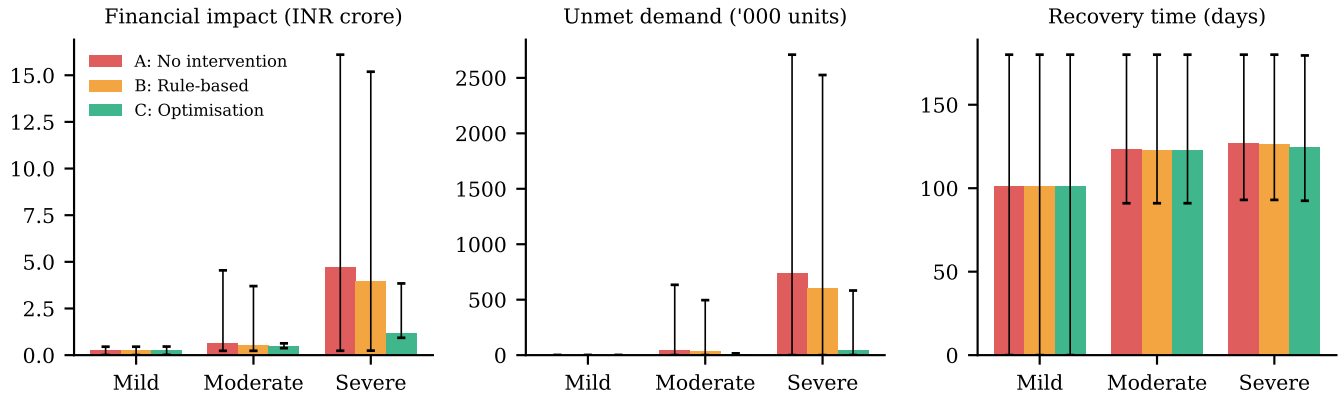


Fig. 3. Nine-scenario outcomes (means and 95 % percentile intervals over 500 replications; simulated INR on synthetic data).

TABLE III
EXP. 2B: GRAPH VS. TABULAR RISK CLASSIFICATION (TEST SPLIT, PLATT-CALIBRATED; MEAN±SD OVER 5 SEEDS). SYNTHETIC: 30-DAY DISRUPTION (PREVALENCE 7.4%); SUPPLYGRAPH (REAL FMCG): 7-DAY SERVICE SHORTFALL (PREVALENCE 28.8%).

Data	Model [graph]	PR-AUC	ROC-AUC	Brier
Synthetic	XGBoost (tabular)	0.122±0.000	0.626	0.0680
Synthetic	MLP (no graph)	0.104±0.011	0.580	0.0685
Synthetic	GCN [observed]	0.104±0.005	0.590	0.0684
Synthetic	GCN [obs.+pred.]	0.111±0.011	0.591	0.0683
Synthetic	GCN [oracle]	0.107±0.004	0.594	0.0684
Synthetic	GraphSAGE [observed]	0.105±0.004	0.588	0.0685
Synthetic	GraphSAGE [obs.+pred.]	0.106±0.007	0.579	0.0685
Synthetic	GraphSAGE [oracle]	0.099±0.006	0.587	0.0685
Synthetic	GAT [observed]	0.100±0.009	0.585	0.0686
Synthetic	GAT [obs.+pred.]	0.098±0.010	0.585	0.0686
Synthetic	GAT [oracle]	0.108±0.005	0.598	0.0683
SupplyGraph	Logistic Regression	0.488±0.000	0.701	0.1841
SupplyGraph	XGBoost (tabular)	0.454±0.000	0.668	0.1905
SupplyGraph	MLP (no graph)	0.475±0.007	0.701	0.1905
SupplyGraph	GCN	0.459±0.030	0.633	0.1961
SupplyGraph	GraphSAGE	0.461±0.017	0.607	0.1975
SupplyGraph	GAT	0.443±0.040	0.660	0.1932

TABLE IV
EXP. 3 (REAL DATACO): LATE-DELIVERY PREDICTION, CHRONOLOGICAL TEST SPLIT (34,554 ORDER LINES, LATE RATE 57.3%), THRESHOLD 0.5.

Model	ROC-AUC	PR-AUC	F1	Brier
Logistic Regression	0.749	0.839	0.682	0.1878
Random Forest	0.777	0.855	0.710	0.1768
XGBoost	0.780	0.858	0.713	0.1768
Shipping-mode rate	0.744	0.785	0.670	0.1881

units unmet (minimum service 64 %, impact 4.72 crore); the rule-based policy reduces this to 605k (3.98 crore); it can exclude only processors with *observed* links to the failed traders and orders reactively, once cover has fallen. Optimisation pre-books de-rated alternates and expedited freight, cutting unmet demand to 39k units with full recovery in 88 % of replications (1.16 crore; recovery procurement spend 4.62 crore). The paired CRN saving of C over A is 3.56 crore (95 % CI 3.25–3.87) and over B 2.82 crore. In the moderate case optimisation saves 16.7 lakh on average over no intervention, but this is tail protection: no intervention is cheapest in 82 % of moderate replications, whereas optimisation is cheapest in 73 % of severe ones. It is not free: in the mild case, where buffers suffice, it over-hedges and costs 0.82 lakh more than doing nothing. The relative standard error of the severe no-intervention mean after 500 replications is 3.9 %, and recovery time is censored at the horizon in at most 3 % of replications.

V. DISCUSSION

Relevance and significance. The system turns predictions into an auditable action and allocation for a real FMCG procurement decision; the value is concentrated in rare, severe network-level shocks, which supports a tiered policy: monitor routinely, escalate to optimisation when the reported shock is large. **Originality.** The contribution is the leakage-controlled integration and the measurement of what hidden links cost, not a new learner; our negative GNN findings are informative for practitioners with sparse supplier maps. **Limitations.** Supplier-level results rest on synthetic data whose generator we designed; performance on real supplier networks is not established. Economic parameters are illustrative. Policy C uses predicted links whose quality is moderate. The assistant's Gemini function-calling path was executed with live tool calls but is throttled on the free tier (5 requests/min), so a deterministic tool router answers when Gemini is busy. No real-time deployment is claimed.

VI. REPRODUCIBILITY

Code, synthetic data, processed SupplyGraph data, configurations and logged runs: https://github.com/SamarthKamat/AIML-PROJECT (branch `claude/exciting-euler-5v1r2l`). `bash run_all.sh` downloads DataCo and SupplyGraph at pinned commits with hash checks, regenerates data (seed 42), runs Exps. 1–5 (GNN seeds 0–4), verifies 61 reconciliation checks (confusion matrices, split counts, prevalence, recomputed metrics, CRN integrity, impact components), rebuilds every table of this paper from result files, and runs 35 tests (unit and API, on SQLite and MySQL). Software: Python 3.13,

TABLE V

NINE-SCENARIO EVALUATION (MEANS OF 500 COMMON-RANDOM-NUMBER REPLICATIONS; SIMULATED INR ON SYNTHETIC DATA). SEVERITY = CAPACITY LOSS OF THE FIVE LARGEST TIER-2 COCOA-BEAN TRADERS FOR A NOMINAL 60 DAYS (LOG-NORMAL). POLICY A NONE, B RULE-BASED, C OPTIMISATION. DISRUPTED CAPACITY (TIME-AVERAGED OVER THE HORIZON) IS IDENTICAL ACROSS POLICIES WITHIN A SEVERITY.

Scen.	Sev.	Pol.	Aff. sup.	Aff. mat.	Unmet (k units)	Inv. short. (L)	Procure. (L)	Transport (L)	Impact, Cr [95% PI]	Recov. d	Alt. util. %	Full rec. %
S1	Mil.	A	10.4	2.4	0.0	0.0	0.0	0.00	0.26 [0.00, 0.45]	101	0	100
S2	Mil.	B	10.4	2.4	0.0	0.0	0.0	0.00	0.26 [0.00, 0.45]	101	1	100
S3	Mil.	C	10.4	2.4	0.0	0.0	4.8	0.12	0.26 [0.01, 0.46]	101	5	100
S4	Mod.	A	17.1	3.9	47.6	2.8	0.0	0.00	0.61 [0.23, 4.54]	123	0	82
S5	Mod.	B	17.1	3.9	29.8	1.9	3.0	0.12	0.51 [0.23, 3.70]	123	16	88
S6	Mod.	C	17.1	3.7	2.6	0.2	86.2	2.95	0.45 [0.38, 0.63]	123	16	97
S7	Sev.	A	19.8	4.0	734.1	46.3	0.0	0.00	4.72 [0.23, 16.10]	127	0	14
S8	Sev.	B	19.8	4.0	605.3	38.9	18.3	0.73	3.98 [0.24, 15.19]	126	42	15
S9	Sev.	C	19.8	3.7	39.2	5.3	461.9	24.20	1.16 [0.93, 3.84]	125	23	88

scikit-learn 1.9, XGBoost 3.4, PyTorch 2.15, PyG 2.8, PuLP 2.9. **AI use declaration.** Claude Code (Anthropic) was used as a programming assistant for code scaffolding, debugging and test generation, and for language editing and checking of bibliographic details. Google Gemini is used only at run time inside the application's assistant, where it may phrase answers solely from tool outputs. All research design choices, experiments, results and interpretations were reviewed and verified by the authors, who take full responsibility for the content of this paper.

VII. CONCLUSION AND FUTURE WORK

Calibrated prediction alone rarely justifies action at supplier level when inventory buffers are healthy; the decisive value lies in exposing hidden dependencies and optimising recovery for severe shocks, where simulated impact fell by 75 %. Future work: validate on real supplier maps (e.g., bills of lading or UN Comtrade cocoa flows), learn risk-aware policies that avoid over-hedging, and integrate verified link feedback into training.

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